

Computer Programming II (CSE2035)

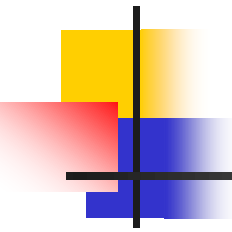
Week 4 Lab Session:

Enum, Structure, and Union Types

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Homework 4-1: Student List

- You will write two functions for managing student's attendance status.
- Let's use the following struct to store student information in a linked list (**do not change these type definitions; you can only touch student-list.c file**)

```
typedef enum { Present, Late, Absent } Attendance;  
  
typedef struct StudentNode {  
    char name[MAX_NAMELEN + 1];  
    Attendance state;  
    struct StudentNode *next;  
} StudentNode;
```

Homework 4-1: Student List

■ Function Prototype

```
void addStudentAtEnd(StudentNode **phead, char *name,  
                    int hour, int minute);
```

■ Create a new node for the linked list and fill in the student information in it. Attendance state is 'Present' if one came before 9:00, 'Late' if one came between 9:00~9:15, 'Absent' if one came after 9:15

- Let's consider 9:00 as 'Present' and 9:15 as 'Late'

■ Assumption

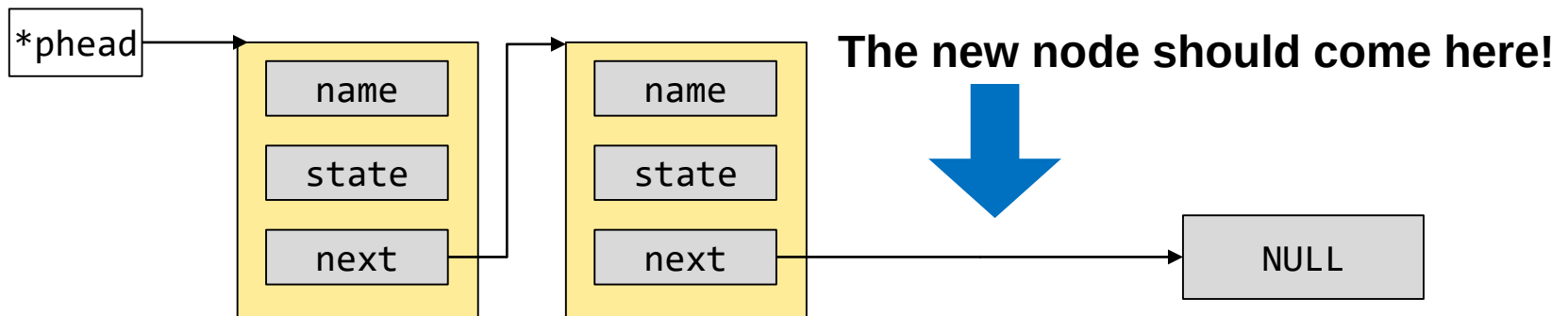
- name is a valid string. If its length is longer than MAX_NAMELEN, copy the first MAX_NAMELEN characters and append '\0'.
- The range of hour is [0, 23]
- The range of minute is [0, 59]

Homework 4-1: Student List

■ Function Prototype

```
void addStudentAtEnd(StudentNode **phead, char *name,  
                    int hour, int minute);
```

- After creating a node, **append it at the end** of the list. Argument 'phead' is a double pointer, where '*phead' points to the first node of the linked list.
- When the linked list is empty, then '*phead' is NULL. In that case, you must update *phead with the first node you created.



Homework 4-1: Student List

■ Function Prototype

```
void printStudentList(StudentNode *head);
```

■ Iterate through the linked list (which is provided as 'head') and print out the state of each student

■ Output must be in the following format

- If the format doesn't match, my automated system will grade your answer as wrong
- Note the order of the printed students

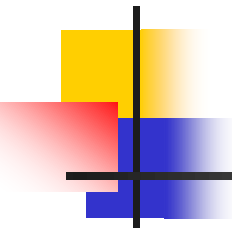
Input data

```
Alice 9 10
Brian 8 50
Chris 9 38
```

Output message

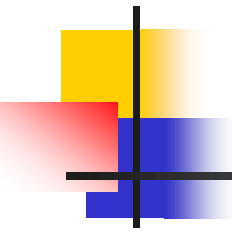
```
Alice is late.
Brian is present.
Chris is absent.
```

'\n' at
the end

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Submission Guideline

- **Submit the completed `student-list.c` file to the “Week 4 Lab Assignment” page in cyber campus**
 - Remember that you are allowed to change only `student-list.c`
 - Do not zip the directory, just upload this file in the form
 - If your submission format is wrong, 10% deduction
 - If the submitted file does not compile with the "make" command, cannot give you any point
- **Deadline is 9/28 (Wednesday) 23:59**
- **Late submission is allowed until 9/30 (Friday) 23:59**
 - 20% deduction

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Q & A Session
